

ALOK RANJAN

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PROFESSIONAL SUMMARY

M.Tech (Propulsion) candidate at IIT Kanpur with a strong foundation in fluid mechanics, heat transfer, and CFD analysis, built through hands-on solver development, multiphase/EHD spray simulation, and turbulence modeling (RANS/LES). Experienced in developing and validating digital prototypes against physical/experimental benchmarks, and comfortable working across commercial (ANSYS Fluent) and open-source (OpenFOAM) simulation environments on HPC systems. Seeking to apply analytical and simulation expertise to mature digital-prototype (DP) capabilities in flow and thermal domains.

TECHNICAL SKILLS

- Core Domains:** Computational Fluid Dynamics (CFD), Heat Transfer, Fluid Mechanics, Turbulence Modeling (RANS/LES), Multiphase & Spray Dynamics, 1D/Reduced-Order Modeling, Parallel Computation
- Simulation Tools:** ANSYS Fluent, OpenFOAM, ICEM CFD, Gmsh, ParaView
- Solvers & Computing:** PETSc, OpenMPI, HPC (Linux cluster environments), Python
- Design/FE Tools:** SolidWorks, CATIA, Autodesk Inventor, ABAQUS, LaTeX

RESEARCH EXPERIENCE & PROJECTS

M.Tech Thesis: Numerical Investigation of Electrostatic Spray Dynamics | IIT Kanpur | Guide: Dr. Ashoke De | Jul 2025 – Present

- Developed sprayElectroFoam, a custom OpenFOAM solver implementing electrohydrodynamic (EHD) coupling, Lagrangian droplet tracking, and image-charge force modeling.
- Implemented RANS and LES turbulence models on HPC to analyze transient spray plume dynamics and droplet stability — directly applicable to flow/thermal digital-prototype development.
- Validated droplet morphology and simulation predictions against experimental benchmarks using Python, reinforcing a test-driven approach to maturing analytical models.
- Managed end-to-end CFD workflows: geometry preparation, meshing, parallel solver execution, and statistical post-processing of high-density datasets.

B.Tech Thesis: Failure Analysis of Bio-Inspired Self-Similar Hierarchical Composites | IIST Thiruvananthapuram | Guide: Dr. Anup S | Jan – May 2022

- Performed finite element simulations in ABAQUS to study strain fields and stress distribution, and computed stiffness via computational and analytical methods.
- Designed and fabricated hierarchical composite specimens via 3D printing and conducted tensile tests to validate simulation predictions against physical test data.

Summer Internship: CFD Analysis of 2D Flow Past a Square Cylinder | IIST Thiruvananthapuram | Guide: Dr. Manu K V | Jun – Oct 2021

- Simulated unsteady flow and vortex shedding at low Reynolds numbers using OpenFOAM.
- Computed lift/drag coefficients and Strouhal numbers, validating simulation results against benchmark experimental data.

Course Project: Aerospace Vehicle Design | Mentored by Dr. Ayyappan G (VSSC, ISRO) and Dr. Manoj T Nair (IIST, ISRO) | Aug – Nov 2021

- Designed and optimized inflatable aerodynamic decelerator (IAD) geometry and parachute systems; developed trajectory analysis codes.

Course Project: Parallel Computation for Sparse Matrix Systems | IIT Kanpur | Instructors: Dr. Ashoke De, Dr. Malay K Das | Jan – May 2025

- Implemented serial, OpenMP, and MPI solvers for sparse matrix systems; benchmarked performance and scalability using PETSc.

EDUCATION

Year	Degree	Institute
2024 – Present	M.Tech, Aerospace Engineering (Propulsion)	Indian Institute of Technology (IIT) Kanpur
2018 – 2022	B.Tech, Aerospace Engineering	Indian Institute of Space Science and Technology (IIST), Thiruvananthapuram
2017	Senior Secondary	Prayas Boys Residential School, Raipur
2015	Secondary	Loyola Higher Secondary School, Kunkuri

RELEVANT COURSEWORK

- M.Tech:** Fluid Dynamics, Heat Transfer, Compressible Flow, Combustion, Computational Methods, Numerical Linear Algebra, Parallel Computation for Sparse Matrix Systems
- B.Tech:** Fluid Mechanics, Heat Transfer, Air-Breathing Propulsion, Thermodynamics, Optimization Techniques in Engineering, Industrial Engineering, Principles of Management Systems

ACADEMIC ACHIEVEMENTS

- Qualified GATE 2024 (Aerospace) and selected for M.Tech Propulsion at IIT Kanpur.
- Selected for B.Tech in Aerospace Engineering at IIST Thiruvananthapuram through JEE Advanced (2018).